

BanjoMaster Visualizer: Multimodal Banjo Finger Position & Tab App

BanjoMaster Visualizer is an interactive web application designed to translate multimodal inputs—**Voice commands**, **Text input**, and **Sheet Music photos/scans**—into real-time visual finger positions on a 5-string banjo fretboard and tablature.

1. Key Features & Capabilities

- **Authentic 5-String Banjo Geometry:**
 - Standard **Open G tuning (gDGBD)**: 1st string = D4, 2nd string = B3, 3rd string = G3, 4th string = D3, 5th string = G4.
 - Accurate short 5th string: starts physically at the **5th fret** with its tuning peg and pip. Frets 1–4 are inaccessible on the 5th string.
 - Nickel-silver frets 0 through 15 with pearl position marker inlays at frets 3, 5, 7, 10, 12 (double dot), and 15.
- **Visual Finger Position Mapping:**
 - Color-coded finger bubbles showing standard bluegrass/folk finger assignments:
 - **1 (Index)**: Blue
 - **2 (Middle)**: Green
 - **3 (Ring)**: Amber
 - **4 (Pinky)**: Purple
 - **T (Thumb)**: Magenta / Goldenrod
 - **O (Open String)**: Cyan badge at nut
 - **X (Muted String)**: Red badge at nut
 - Instant toggle between **Finger Numbers** and **Note Pitch Names** (G, B, D, F#, etc.).
- **Multimodal Input Engines:**
 1. **Voice Input (Web Speech API)**: Speech recognition listening for conversational commands like *"G chord"*, *"C major"*, *"D seventh"*, *"E minor"*, *"Forward roll"*, *"Notes G B D"*, or *"Clear fretboard"*.
 2. **Text Input & Chord Browser**: Direct text search box with instant autocompletion, plus quick-selection grids for common open chords (G, C, D, D7, Em, Am, F, B7, G7) and movable up-the-neck shapes (Barre, F-shape, D-shape).
 3. **Sheet Music / Photo Input (Canvas OMR)**: Drag-and-drop or camera upload for sheet music images, with automatic staff line detection (Treble clef), notehead blob localization, pitch resolver (Middle C4 up to A5), and an interactive confirmation canvas where users can double-click to add notes, delete artifacts, or adjust note pitches before mapping them to the banjo neck.
- **Acoustic Banjo Sound Synthesizer (Web Audio API)**:
 - Custom acoustic banjo physical modeling synthesis (pluck attack transient, skin head bandpass filter resonance at ~1800 Hz, fast metallic decay).
 - One-click **Strum Chord** brush.

- Scruggs 3-finger roll pattern playback (Forward Roll, Backward Roll, Alternating Thumb Roll) with tempo control (60 to 180 BPM).
- **Banjo Tablature (TAB) Engine:**
 - Generates standard 5-line banjo tablature reflecting the active chord or notes in real time.
- **Alternative Tunings:**
 - Includes Open G (gDGBD), Double C (gCGCD), Drop C (gCGBD), and D Tuning (f#DF#AD).

2. Voice Commands Reference

Voice Command	Recognized Intent	Banjo Fretboard Action
"G chord" or "Play G major"	Chord Selection	Shows Open G [0, 0, 0, 0, 0] with all open strings
"C major" or "C chord"	Chord Selection	Frets 2 (String 1), 1 (String 2), 0 (String 3), 2 (String 4), 0 (String 5)
"D seventh" or "D7"	Chord Selection	Frets 0 (String 1), 1 (String 2), 2 (String 3), 0 (String 4), 0 (String 5)
"Notes G B D"	Melodic Note Mapping	Locates first-position frets for pitches G, B, and D
"Forward roll"	Picking Pattern Playback	Plays Scruggs 8-note forward roll (Strings 3-2-1-5-2-1-5-1) over active chord
"Clear fretboard"	Reset	Clears all frets and mutes all strings

3. Optical Music Recognition (OMR) Workflow

1. **Upload Image:** Select an image file or drag-and-drop a photo/scan of a single treble clef staff measure into the upload zone.
2. **Built-in Samples:** Alternatively, click one of the preset buttons (*C Major Triad*, *G Banjo Arpeggio*, or *"Cripple Creek" Lick*).
3. **Visual Verification & Editing:** The interactive canvas renders the staff lines and detected notes. Double-click anywhere on the staff to add a missing note. Click the red 'X' next to

any note pill in the detected notes strip to remove artifacts.

4. **Map to Banjo:** Click *"Map Sheet Notes to Banjo Fretboard"* to translate the musical pitches to optimal string and fret finger positions.

4. Running the Standalone Application

The entire application is completely self-contained in a single file with zero external dependencies. It uses native HTML5, SVG, Web Speech API, Canvas API, and the Web Audio API.

To run the app on any computer or mobile device, save the application code as `index.html` and open it in Google Chrome, Microsoft Edge, Safari, or Firefox.